

From model to ROS 2 system

A trained model becomes useful when preprocessing, runtime, messages, timing, geometry, and evaluation agree on explicit contracts.

15–20 MINUTES

Course synthesis

PIPELINE DIAGNOSIS

1 • Perceive

Acquire synchronized sensor data.

2 • Infer

Prepare tensors and run the model.

3 • Interpret

Decode and validate outputs.

4 • Communicate

Publish typed ROS 2 messages.

ROS 2 provides communication, not perception

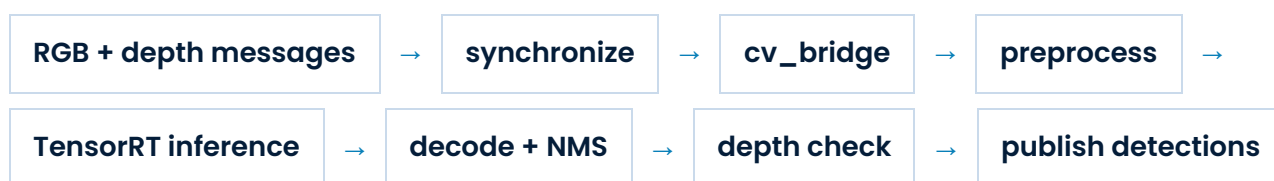
A ROS 2 **node** is a process-level participant in a graph. A **topic** carries a stream of typed messages from publishers to subscribers. The message type defines the data structure. Quality of Service (QoS) settings define delivery behavior such as reliability and queue history.

ROS 2 does not train the model or understand images. It transports sensor observations and detection results, supplies configuration and lifecycle conventions, and connects perception to other robot components.

Central mental model

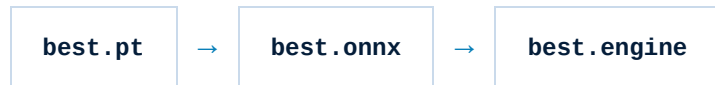
Every arrow in a robot vision pipeline is a contract: type, shape, units, coordinates, timing, policy, and evidence.

The complete runtime path



Synchronization pairs observations close enough in time for the application. **cv_bridge** converts between ROS image messages and OpenCV arrays while respecting encoding. The model runtime executes a fixed artifact. Post-processing maps raw outputs to the project's detection contract. Publishers then expose standard, vendor-specific, and debug representations.

The artifact path



The PyTorch checkpoint is the training-framework artifact. ONNX expresses a portable computation graph and named tensor interfaces. TensorRT builds a target-optimized engine for the Jetson and selected precision. Conversion introduces a new boundary, so each artifact needs structural and behavioral checks. A successful load or smoke inference establishes execution under tested conditions, not deployment accuracy.

A diagnostic method

1. **State the symptom:** for example, no final detections in live room frames.
2. **Locate the last trusted boundary:** messages arriving, tensor valid, candidates present, geometry kept, output published.
3. **Form one hypothesis:** such as domain shift causes class scores below threshold.
4. **Predict evidence:** lowering the threshold should reveal correctly located low-score candidates if that hypothesis is true.
5. **Run a controlled test:** hold other variables fixed and record inputs, outputs, versions, and thresholds.
6. **Update the conclusion:** distinguish confirmed result from remaining alternatives.

Browser exercise: localize a failure

RGB and depth topics publish at expected rates. The callback runs. TensorRT reports candidates, but none cross the class-score threshold. Which boundary should you investigate first?

Reveal the next investigation

Second scenario: valid boxes exist before geometry, but all disappear afterward. Inspect depth alignment, units, validity, and geometry policy before retraining the detector.

Applied to our cubes

REPOSITORY BEHAVIOR `_sync_cb` converts synchronized RGB/depth images, runs `_infer`, applies geometry decisions, and publishes outputs. The node provides standard `vision_msgs`, vendor objects, and a debug image.

CURRENT STATUS M5 is PARTIAL: the runtime path and empty-scene behavior work, while reliable positive detection in the room remains unresolved.

CONTROLLED NEXT EXPERIMENT Build a labeled, capture-session-separated room set with all three cube colors and empty scenes. Compare the current and candidate checkpoints at fixed preprocessing and thresholds using class-and-IoU matching, then repeat a live run only if offline gates pass.

Project anchors: architecture, milestones, runtime node, and artifact chain.

Final check

What is the strongest first response to a pipeline failure?

- ☐ Retrain immediately with more epochs
- ☐ Locate the failing contract boundary
- ☐ Replace every middleware message type

Explain it back: course synthesis

Give a 90-second walkthrough from camera observation to evaluated ROS 2 detection. Include image contract, learned features, inference, boxes and scores, depth, message publication, one metric, and one current limitation. Then propose the smallest experiment that would reduce uncertainty.

Primary sources

- ROS 2 Humble: Topics and QoS.
- ONNX introduction and TensorRT Quick Start.